

How a school swarms

collective motion for the aquarium level's ten followers · the Couzin zone model, measured against our own Forth implementation

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is $13.4 \times 3.4 \times 4.7$ BL inside. A **school** is polarised (p_group near 1), a **swarm** is not: moving the zone width Δr_o switches between them.

A · Four collective states from one rule, simulated here (N = 100, the paper's $r_r, \alpha, \theta, s, \sigma$)

A · swarm



top view

B · torus



spin-axis view

C · dynamic parallel



side view

D · highly parallel



side view

$\Delta r_o = 0, \Delta r_a = 15$
p_group 0.16 · m_group 0.10

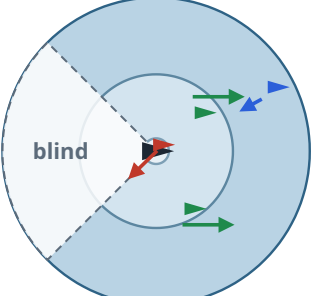
$\Delta r_o = 1, \Delta r_a = 12$
p_group 0.09 · m_group 0.90

$\Delta r_o = 3, \Delta r_a = 10$
p_group 0.90 · m_group 0.03

$\Delta r_o = 10, \Delta r_a = 10$
p_group 1.00 · m_group 0.02

ours · N = 100 simulated here; the zone widths are ours, the paper prints none

B · The model: three zones and a blind volume



repulsion zone, $r_r = 1$
anyone here: turn away, nothing else counts

orientation zone, width Δr_o
those here: match their heading

attraction zone, width Δr_a
those here: turn toward them

blind volume, 90° behind
 $\alpha = 270^\circ$: cannot see here

Drawn at the school setting: $\Delta r_o = 5, \Delta r_a = 6$

verified · zones, α, r_r : Couzin 2002

ours · widths drawn

C · The map: p_group over the two zone widths

attraction width Δr_a

orientation width Δr_o (BL)

0 1 2 3 4 6 8 10 12 15

15	S	T	?	D	D	P	P	P	P	P	P	P	P	P	P
12	S	T	?	D	D	P	P	P	P	P	P	P	P	P	P
10	S	?	D	D	D	P	P	P	P	P	P	P	P	P	P
8	S	F	D	D	D	P	P	P	P	P	P	P	P	P	P
6	F	F	F	D	D	P	P	P	P	P	P	P	P	P	P
4	F	F	F	D	D	P	P	P	P	P	P	P	P	P	P
3	F	F	F	F	D	P	P	P	P	P	P	P	P	P	P
2	F	F	F	F	D	P	P	P	P	P	P	P	P	P	P
1	F	F	F	F	F	P	P	P	P	P	P	P	P	P	P
0	F	F	F	F	F	F	P	P	P	P	P	P	P	P	P

p_group (alignment)

< 0.25 < 0.6 < 0.9 ≥ 0.9

S swarm: low p, low m

T torus: low p, high m

D dynamic parallel: high p

P highly parallel: $p \approx 1$

F fragments (more than half)

? seeds disagree: no one state

our swarm, torus, school

ours · N = 100, 3 replicates, 700 steps (paper: 30)

D · The rule in eight lines, and the Forth word that does each

1 See the others, except behind

2 Repulsion if anyone is within r_r

3 Else orientation and attraction

4 Nobody in any zone: keep the heading

5 Turn toward d by at most $\theta\tau$, move s τ

6 The leader counts w times in sums 3

7 Walls repel like a neighbour, and are a hard limit

8 A dart startles the followers (not called yet)

$u_{ij} \cdot v_i \geq \cos(\alpha/2)$

$d = -\sum u_{ij}$ (nothing else counts)

$d_o = \sum v_j, d_a = \sum u_{ij}; d = (d_o + d_a)/2$

$d = v_i$

$v \leftarrow \text{rotate}(v, d, \theta\tau); c \leftarrow c + s \tau v$

$\sum w_j v_j, \sum w_j u_{ij}$

within 0.6 BL: one more repulsion; past the box: put back, heading reflected

swim away from the leader for 0.5 s

sch-pair

sch-pair

sch-pair, sch-want

sch-want

sch-turn, sch-follow

sch-pair (weight)

sch-wall, clamp-axis

sch-startle-all

The paper's random error ($\sigma = 0.05$ rad) is left out of the Forth, and the game adds none. Our units differ from the paper's, on purpose: at its $s = 3$ BL/s and $\theta = 40^\circ$ /s a 90° turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours: $s = 2$ BL/s, $\theta = 120^\circ$ /s, 1.50 BL, radius 0.95 BL.

verified · rules 1–5 and the paper's s, θ : Couzin 2002, eqns 1–3 and Fig. 3

ours · rules 6–8, our s, θ , the 0.6 BL wall zone, the 0.5 s startle

F · The Forth core, school.fth: the four rule words, and what they cost

sch-pair	168 B · 24 lines	sch-want	88 B · 11 lines	sch-turn	104 B · 12 lines	sch-follow	131 B · 17 lines
one neighbour: blind volume, then repulsion, orientation or attraction : sch-pair (j --) MB_YOU sc! MB_X MB_U you-delta MB_Y MB_U 1 + you-delta MB_Z MB_U 2 + you-delta MB_U vlen2 dup 0.000001 > over reach2 < & if sch-sqrt dup MB_DIST sc! 1 swap / MB_U vscale heading-dot-u MB_COSB par@ >= if r-rep inside if -1 MB_SUM_R acc+u MB_N_R count+ else r-ori inside if you-weight MB_SUM_0 acc+v MB_N_0 count+ else you-weight MB_SUM_A acc+u MB_N_A count+ then then then else drop then ;	the wanted direction from the three sums, or the old heading : sch-want MB_WANT vzero MB_N_R seen? if MB_SUM_R add-wanted else MB_N_0 seen? if MB_SUM_0 add-wanted then MB_N_A seen? if MB_SUM_A add-wanted then then MB_WANT vnorm 0.000001 > not if MB_WANT heading>sc then ;	turn toward it by at most $\theta\tau$, keep the heading unit length : sch-turn MB_VX me sch@ MB_WANT sc@ * MB_VY me sch@ MB_WANT 1 + sc@ * + MB_VZ me sch@ MB_WANT 2 + sc@ * + dup MB_COST par@ >= if drop else perp-all MB_PERP vnorm 0.000001 > not if perp-fallback then turn-all MB_WANT vnorm drop then ;	one follower, start to finish: scan, walls, startle, turn, move : sch-follow MB_ME sc! clear-sums sch-n 0 do i me <> if i sch-pair then loop sch-walls sch-want sch-startle? if sch-startle-away sch-startle-tick then sch-turn MB_WANT sc@ MB_NVX me sch! MB_WANT 1 + sc@ MB_NVY me sch! MB_WANT 2 + sc@ MB_NVZ me sch! 0 advance 1 advance 2 advance clamp-all ;				

3,216 B

of the 64 KB dictionary (4.9 %), 202 code lines; 7.5 ms a step on the Chromecast; 1e-03 off couzin.py

7.0 → 11.3 ms

one tick, 11 fish, Chromecast HD (an earlier 2,936 B version): the bare interpreter, then in the engine

4.2 → 0.28 μ s

per mailbox call: three debug streams the engine ran on every call, found by this timing; 20 fps before

59.9 fps

the real level with ten followers, Chromecast HD, unprofiled; p90 33.4 ms, Director 6.4 ms (profiled)

ours · our measurements: the bare interpreter, and a bench level (the Director runs school.fth every tick) on the real Chromecast, release build

G · The Forth, running in the tank: swarm, torus, school, and a startle

swarm setting (0, 10)



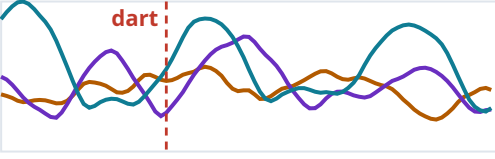
torus setting (1, 10)



school setting (5, 6)



Startle: distance to the leader



leader weight w

dist · align · p · in box

w = 1 2.9 BL · -0.03 · 0.30 · yes

w = 3 2.5 BL · +0.03 · 0.41 · yes

w = 6 2.3 BL · +0.11 · 0.45 · yes

leader weight w

dist · align · p · in box

w = 1 2.8 BL · +0.12 · 0.59 · yes

w = 3 2.3 BL · +0.25 · 0.59 · yes

w = 6 2.1 BL · +0.20 · 0.57 · yes

leader weight w

dist · align · p · in box

w = 1 3.8 BL · +0.03 · 0.86 · yes

w = 3 3.8 BL · -0.00 · 0.85 · yes

w = 6 3.1 BL · +0.09 · 0.78 · yes

In the real level, on the engine (scripts/analyse-aquarium-school.py): resting p_group 0.39, swimming 0.77 (it was 0.24 before a setup bug was fixed), followers aligned with the leader +0.40.

ours · the Forth core in the tank's real box (w = 3, seed 1 drawn; the table is the mean of 3 seeds; "no" = a fish strayed more than 0.5 BL out)

Sources · live links in the PDF; what could not be verified is marked

S1 Couzin et al. 2002, J. Theor. Biol.
verified zone model, states, metrics, Fig. 3

S2 Couzin et al. 2005, Nature
unverified informed minority guides a group

S3 Pitcher 1983, via Wikipedia
unverified shoal to school: a continuum

S4 Ocellaris clownfish (Florida Museum)
unverified site-attached: not schooling fish

S5 the aquarium level and this repository
ours tank, fish, zForth, our runs

How to read the chips
verified the source was opened; the number is on its page
unverified from a summary, or not opened
ours our own choice, maths or measurement

Not claimed: that real clownfish school (they do not), that our sweep shows the paper's hysteresis (it is too coarse), or that the school is tuned: it is in the game, but the lead is weak (alignment +0.40) and the dart does not startle it yet. Data dated 2026-10-01.